# Git: Getting Started

2026-01-05

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### Installation & Configuration

- Git - version control software

> **Note**
>
> MacOS usually comes with `git` pre-installed, so there is no need to install anything else.

- After installing `git`, please configure it: Getting Started - First-Time Git Setup
- The setup steps above should create a `~/.gitconfig` file that has some barebones configuration options. We can expand that a bit by manually editing it with the following content:

```
[user]
    name = your first and last name
    email = yournetid@duke.edu
[alias]
    lg = log --graph --pretty=format:'%Cred%h%Creset -%C(yellow)%d%Creset %s %Cgreen(%cr) %C
[core]
    editor = nano
    autocrlf = input
    preloadindex = true
```

```
    fileMode = false
[color]
    status = auto
    branch = auto
    interactive = auto
    diff = auto
    ui = auto
[push]
    default = current
    autoSetupRemote = true
[pull]
    rebase = false
[merge]
    ff = no
[init]
    defaultBranch = main
```

- The above configuration will:
  - Set your name and email address for commits
  - Set up a `git lg` alias that will show a nice graph of the commit history
  - Change the default editor for making commit messages from `vi` to `nano`.

> **Tip**
>
> You can change this to something else, like VS Code!

- This is a good tutorial on using `git` within VS Code: Using Git source control in VS Code

### Setup Duke GitLab Account

While GitHub is a very popular public git repository hosting site, we will be using Duke's GitLab server. Please make sure you can log into https://gitlab.oit.duke.edu using your Oath2 NetID authentication.

> **Warning**
>
> This is **not** `gitlab.com`!

> **Caution**
>
> Do **not** edit files in remote git repositories using the GitLab web interface!! Only work with your repository files using your local clone of the repository.

### SSH Key Authentication

We will use SSH keys to authenticate us on GitLab to be able to `clone`/`push`/`pull` from GitLab without needing to always enter your username and password.

To setup an SSH key and add it to your GitLab profile:

> **Warning**
>
> The guide below will reference `github.com`; you want to substitute `gitlab.oit.duke.edu`.

1. Generate a New SSH Key
    - While your private key is most secure by encrypting it with a passphrase, you will need to either enter that passphrase everytime the key is used, or you will need to add it to your local credential manager (i.e., keychain) or `ssh-agent` to keep it unlocked.
    - If you are just using this SSH key for this class, you can use an empty passphrase and not need to worry about needing to deal with the password management.
2. Add your SSH key to your GitLab Profile

> **Warning**
>
> The guide above will reference `gitlab.com`; you want to substitute `gitlab.oit.duke.edu`.

### Git Tutorials

We are going to cover `git` extensively in class, but it can really help to review some tutorials ahead of time to get familiar with the nomenclature and high-level overview of the workflow. If you have never used `git` before, I would recommend reviewing the following tutorials:

- Learn Git: Commands, Tutorials and More
- Git Immersion

- Learn Git Branching
- Think Like (a) Git
- Git Tutorial for Beginners: Learn Git in 1 Hour

More comprehensive `git` documentation can be found at https://git-scm.com/doc.